

Auction for Prominence and Price-Directed Search

Theory and Experiment

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SEARCH FRICTION MOTIVATES PAID PROMINENCE

On digital marketplaces, search is costly, so sellers **pay to be inspected first** — Google search ads, Amazon sponsored products, Expedia featured placement.

- **Research question:** How does creating a **market for prominence** affect prices, allocations, and welfare? Does different **magnitude of the friction** induce different market behavior?
- **Could help:** markets and prices may sort buyers to better-matching products.
- **Could hurt:** money-burning and rent-seeking; pure transfers without changing allocations.

THREE-STAGE GAME



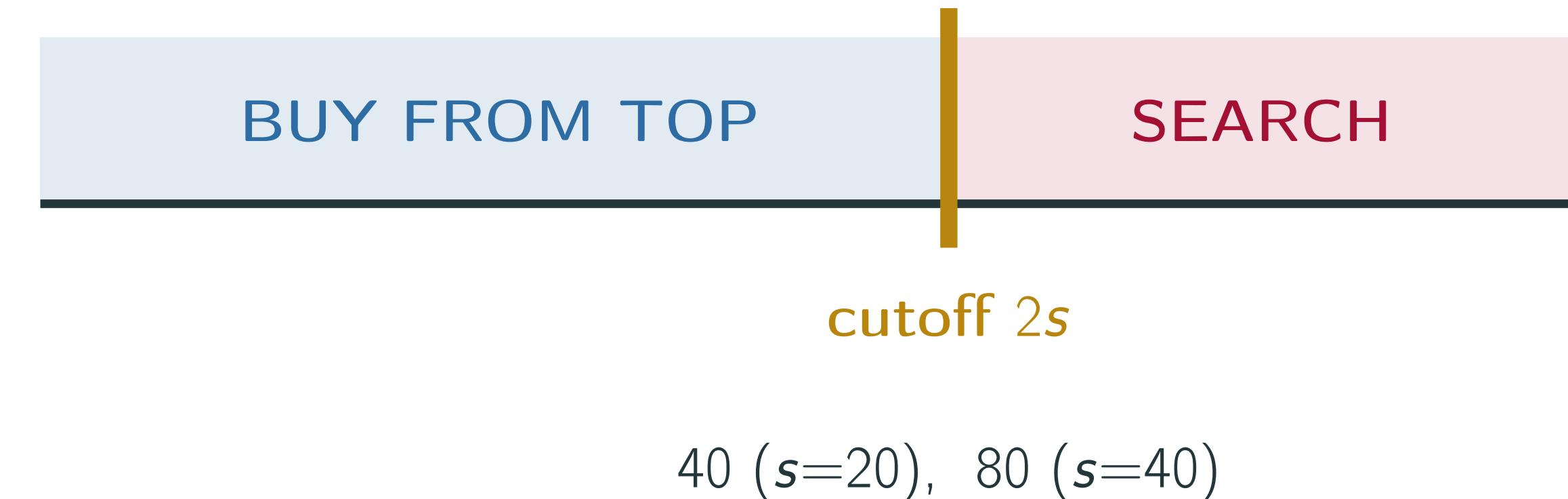
Matches $v_1, v_2 \in \{0, 100\}$, each i.i.d. with probability $q = 0.5$.

Seller payoff = sale revenue — position payment

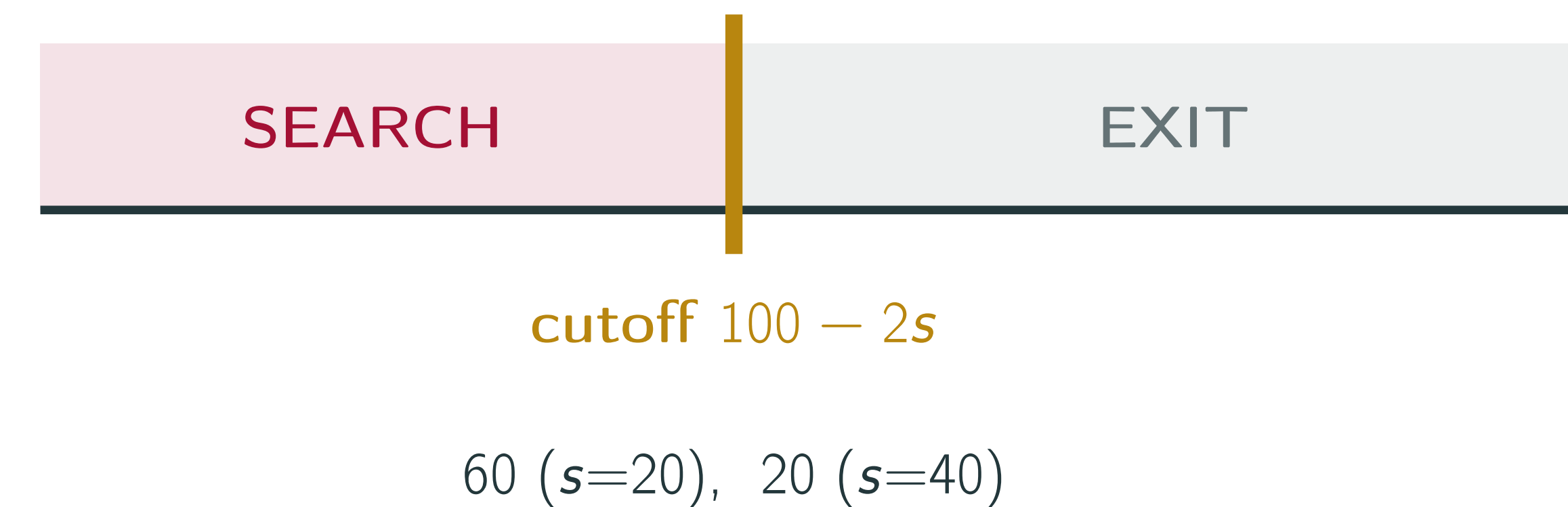
Buyer utility = match value — price — search cost

BUYER'S SEARCH RULE

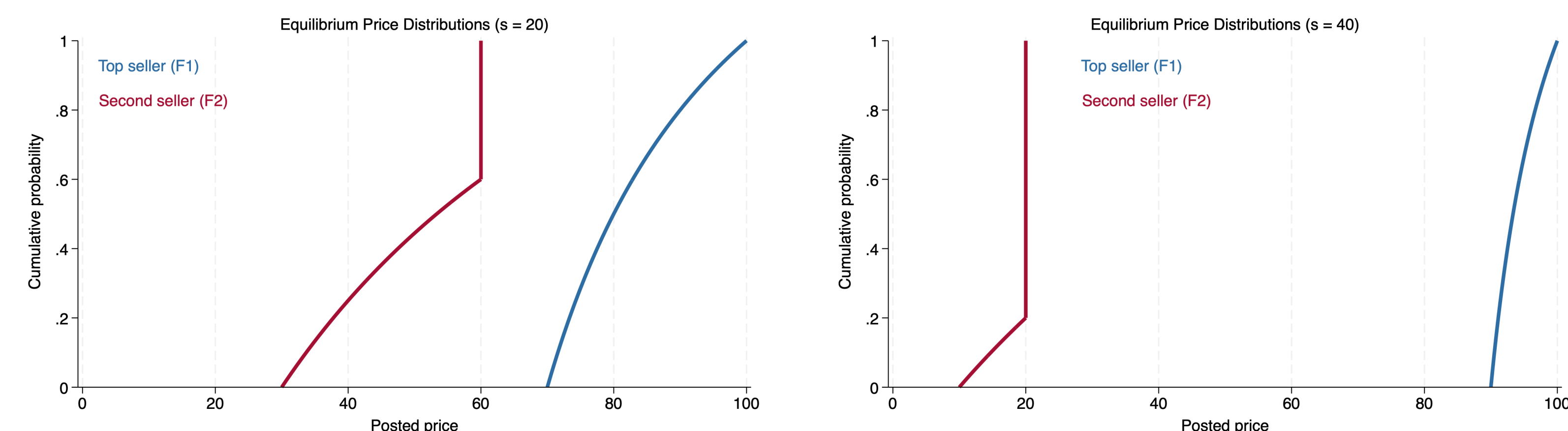
If $v_1 = 100$: search second seller iff $p_1 - p_2 > 2s$



If $v_1 = 0$: search second seller iff $p_2 \leq 100 - 2s$



MIXED-STRATEGY EQUILIBRIUM IN PRICING



- **Price Dispersion:** P_1, P_2 are both dispersed across some ranges.
- **Price Hierarchy:** P_1 first-order stochastically dominates P_2 , for each s .
- **Price Gap (comp. stat.):** $p_1 - p_2$ widens as s increases.

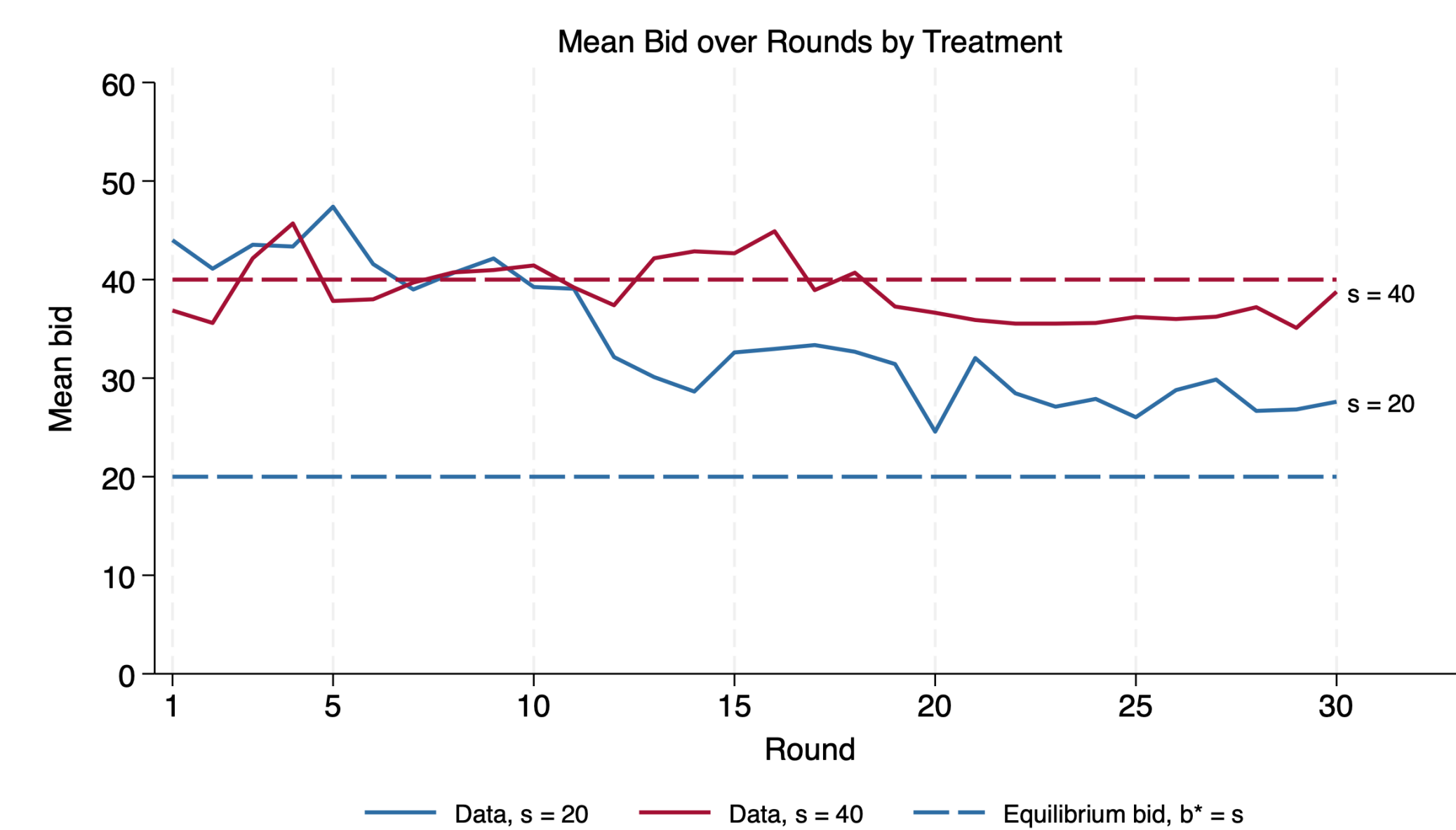
$$b^* = \Pi_1 - \Pi_2 = s$$

Π_k is the continuation revenue of position k and π_k is profit after the position payment. The top seller pays the losing bid $b^{(2)} = s$ and the second seller pays nothing, so equilibrium **equalizes profit across positions**: $\pi_1 = \pi_2$.

EXPERIMENTAL DESIGN

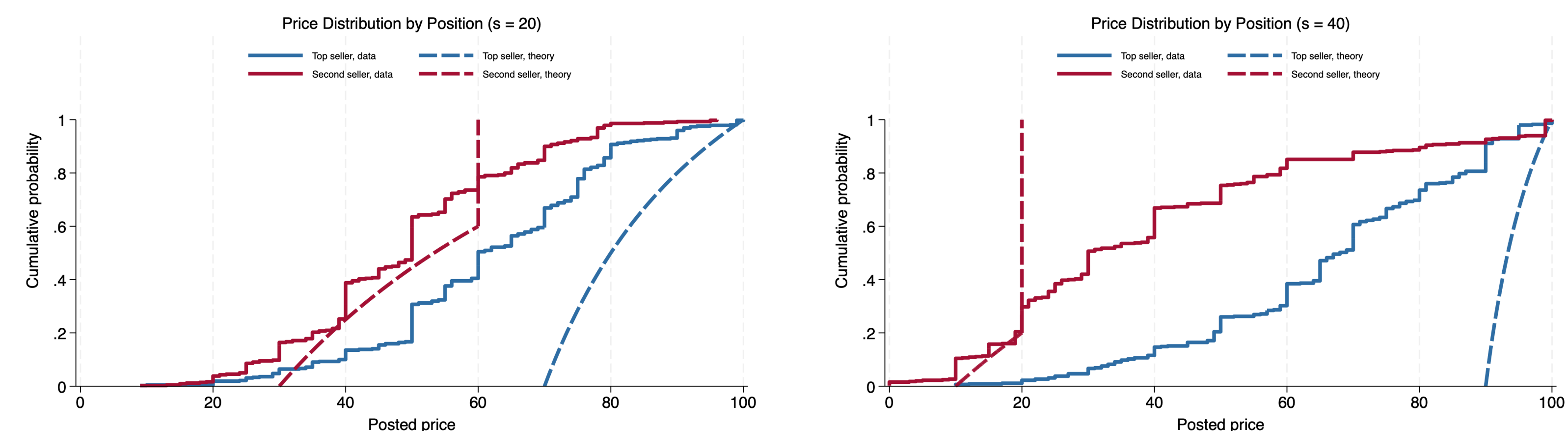
- 6** sessions
- 87** subjects
- 870** markets
- **Each market:** 2 sellers + 1 buyer, three-stage game. Random re-matching across **30 rounds**; match values v_1, v_2 drawn in real time.
 - **Between-subject:** low search cost ($s=20$) vs. high search cost ($s=40$). UVA Veconlab, 3+3 sessions.
 - **Payment:** \$10 show-up + one randomly selected round (10 points = \$1); average \$31.

RESULT 1: SUBMITTED BIDS FOR TOP POSITION



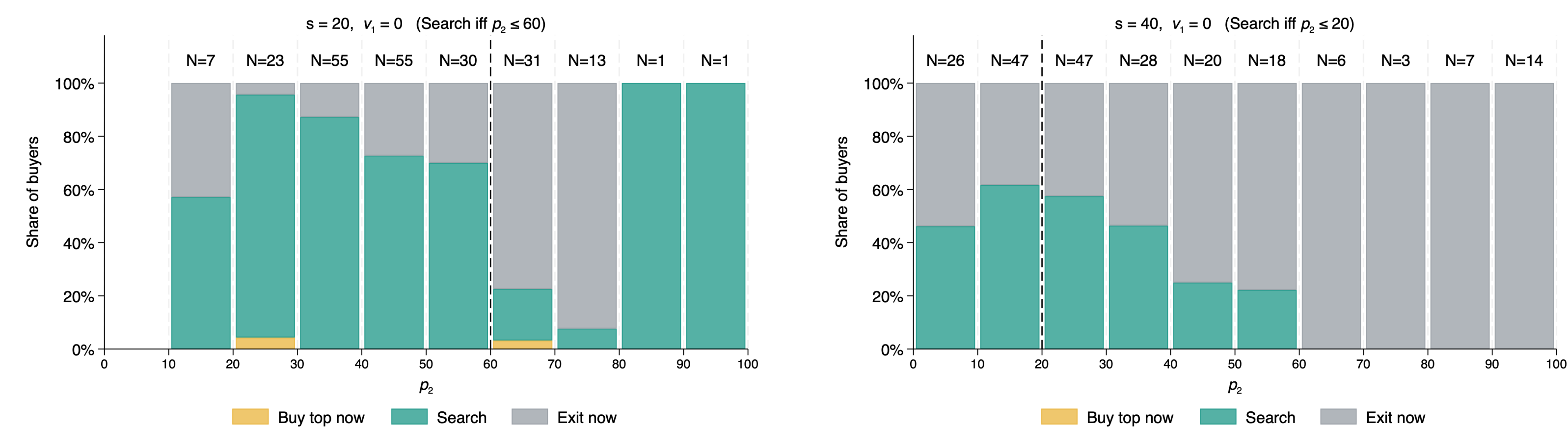
- Over-bidding at $s = 20$ (two-sided t -test p-value $< .001$). All session means above $b^* = 20$.
- Bidding consistent with theory at $s = 40$, with mean bid 38.8 vs. $b^* = 40$ (two-sided t -test p-value = 0.219).
- Bids respond to s (rank-sum test p-value = 0.032) but less than predicted (mean bid 34.0 $\rightarrow$ 38.8).

RESULT 2: PRICE DISPERSION AND HIERARCHY



- Prices far more dispersed than equilibrium (SD: 20.3 vs. 2.8 and 25.4 vs. 2.4 at $s = 40$).
- P_1 stochastically dominates P_2 in both treatments (one-sided KS test p-values $< .001$).
- Price gap compressed in both treatments relative to equilibrium.
- Price gap increases with s , perfect separation by session (rank-sum test p-value = 0.1 at session level).

RESULT 3: SEARCH AND PURCHASE



- After a **good** top match, search follows the theoretical cutoff in both treatments (compliance = 90%), and errors sit at the cutoff: away from it the error rate falls to 2% at $s = 20$ and 8% at $s = 40$.
- After a **bad** top match, compliance falls to 79% at $s = 20$ and 63% at $s = 40$, with under-search 44% below $p_2 = 20$ and over-search 34% above.
- Unlike the good-match case, these errors are **not** confined to the cutoff — the error rate away from it is 19% and 30%.

FINDINGS

- **Over-bidding appears only at the low search cost.** Sellers bid well above the benchmark when search is cheap ($s = 20$), while bidding at $s = 40$ is close to what theory predicts. Willingness to pay for prominence does not scale with the friction that gives prominence its value.
- **Price distributions are distorted, not overturned.** The predicted hierarchy holds — the top seller prices above the second seller in both treatments — but the gap between them is compressed relative to equilibrium, and prices are far more dispersed than theory allows.
- **Search is broadly consistent with the theoretical cutoff.** Buyers follow the cutoff in most cells. The exception is the costly search after a bad top match ($s = 40, v_1 = 0$), where compliance falls and errors run in both directions.